

Project Documentation

Project Title: Job Market Analysis using Microsoft Excel and Power BI

Project Overview

This project analyzes job market data using Microsoft Excel and Power BI to identify salary trends, company insights, industry performance, and employment patterns. The project demonstrates the complete data analysis workflow, including data cleaning, exploratory analysis, dashboard development, and business insights to support data-driven decision-making.

1. Project Objective

The primary objectives of this project are to:

- Analyze job market trends across different industries and sectors.
- Compare salary estimates for various job roles.
- Identify high-paying industries and top-performing companies.
- Visualize key metrics through an interactive Power BI dashboard.
- Demonstrate data analysis and visualization skills using Excel and Power BI.

2. Tools Used

- **Microsoft Excel**
- Power BI

3. Dataset Description

The dataset contains job market information with the following attributes:

Job Title, Job Description, Salary Estimate, Company Name, Industry, Sector, Location, Headquarters Employee Size, Type of Organization, Revenue, Competitors

4. Data Cleaning and Preparation

The dataset was cleaned and prepared in Microsoft Excel before importing it into Power BI.

The preprocessing steps included:

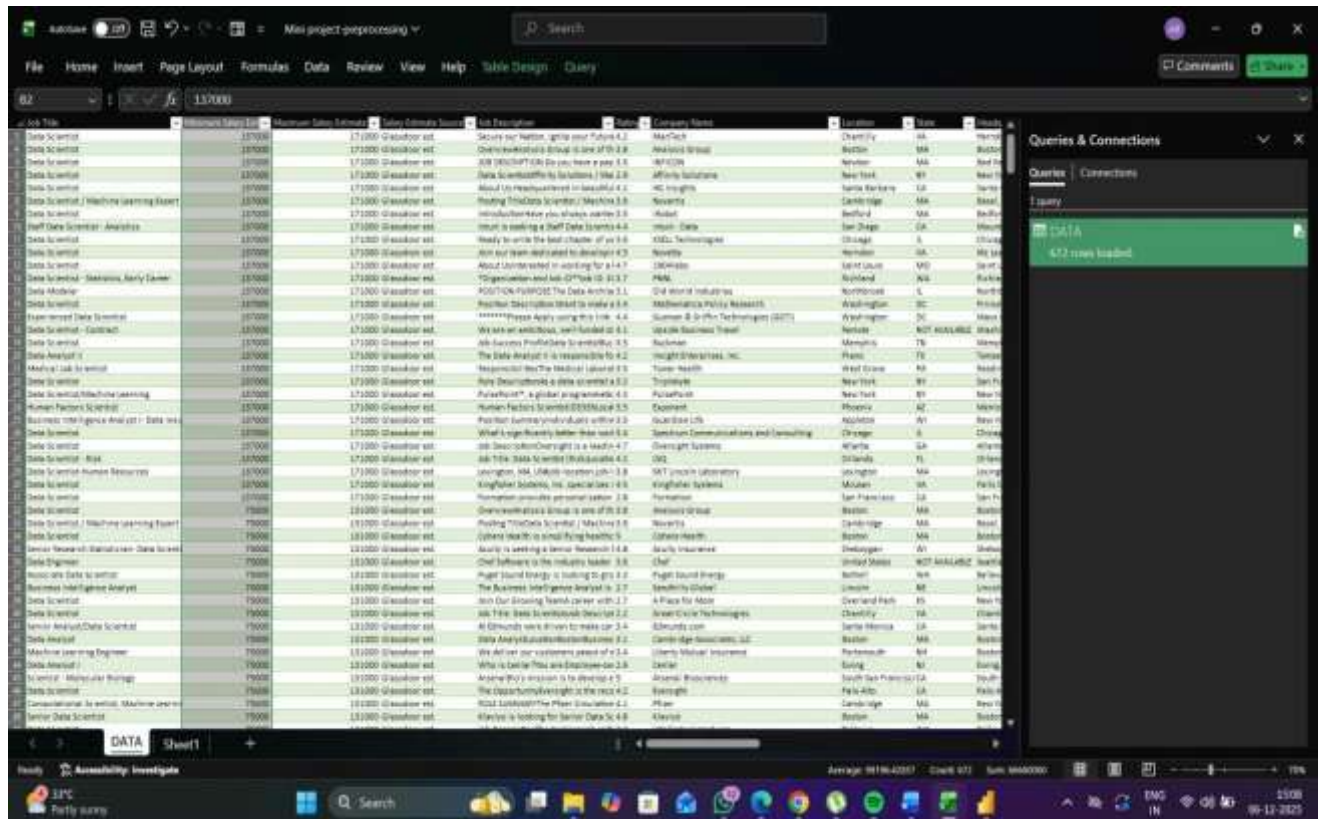
- Removing duplicate records.
- Handling missing or blank values.
- Standardizing text formatting.
- Correcting inconsistent data entries.
- Formatting columns appropriately.
- Preparing the dataset for visualization.

5. Key Insights

- The Information Technology sector has highest revenue (188 USD) than others.
- Among all 406 companies and 169 job roles the maximum and minimum salary of an employee is 331K and 31K respectively.
- The company named Maxar Technologies has the highest number of employees and it is the topest company in rating.
- Highest paying industry is Biotechnology and Pharmaceutical and the lowest is Social assistance.

6.Screenshots

Preprocessed DATA in Excel



Power BI Dashboard



7.Files Included

Uncleaned data

<https://drive.google.com/file/d/1W0KMkHv1a5pTyhu5r9Tt67wvYQTcdthk/view?usp=sharing>

Cleaned data with basic analysis

<https://docs.google.com/spreadsheets/d/1-NIY0Ubck7ZPdlaYyBqhAPdUz6ZvjvK3/edit?usp=sharing&ouid=102583908775806945503&rtpof=true&sd=true>

Power BI dashboard

https://drive.google.com/file/d/1_d3W611ay-znPtXkEm7YW3kZTluMk3Dm/view?usp=sharing

8.How to Use

- Open `DATA.xlsx` to view the cleaned data.
- Open `JOB MARKET.pbi` in Power BI Desktop to explore the visuals.
- Explore the dashboard using the available slicers and filters.